

Dr. B. C. Roy Engineering College, Durgapur

Department of Electrical Engineering
B. Tech (EE) | 5th Semester EE2 | Autonomy

Subject: Power Electronics Subject Code: EE-504

Teaching-Learning Method: Flip Learning (FL)

Date: Tuesday, 1st September 2026 Time: 2:10–3:00 PM Section: EE-2

Faculty: Kingsuk Majumdar, Ph.D. (Engg.) Dept. of EE, BCREC

i Topics: Ideal vs. Practical Switching Characteristics of Power Semiconductor Devices | Turn-On and Turn-Off Times | Switching Losses | Two-Transistor Model of SCR | Regenerative Turn-On Mechanism

Module: 1 (Introduction) and Module 2 (PNPN Devices)

CO: CO1 **Bloom's Level:** BL2 (Understand) for Q1–Q5 | BL3/BL4 (Apply/Analyse) for Q6–Q10

📅 Pre-Class Note: EE504_FL_PreClass_Note_V2 (shared on 30-August-2026, Sunday evening)

📁 What is Flip Learning?

In Flip Learning, **content delivery moves out of the classroom** into the student's own pre-class study time. The class period is used entirely for higher-order application and discussion.

Pre-class (your responsibility before 1st September 2026): Read the study note EE504_FL_PreClass_Note_V2 shared via the class group on 30-August-2026. Estimated reading time: 40–50 minutes. **No theory will be taught in class.**

In-class (1st September 2026, 2:10–3:00 PM):

- **Phase 1 – Individual MCQ (25 min, 2:10–2:35 PM):** Attempt all 10 questions individually. No discussion permitted. Write Name, Roll No., and answers on the response sheet.
- **Phase 2 – Faculty Consolidation (20 min, 2:35–2:55 PM):** Faculty discusses correct answers, common errors, and GATE/ESE tips for each question.
- **Phase 3 – Doubt Clearance (5 min, 2:55–3:00 PM):** Open Q&A on concepts and solutions.

All response sheets must be submitted before leaving. These are mandatory NBA file documents.

 Instructions:

- **10 questions** (Q1–Q5: Conceptual, Q6–Q10: Analytical / Application).
- Each question: **2 marks**. Total: **20 marks**.
- **Negative marking:** $-\frac{2}{3}$ mark per wrong answer. No deduction for unattempted questions.
- Mobile phones must be switched off during Phase 1.
- Write **Name, Roll No.** clearly on the response sheet.
- Show brief working for Q6–Q10 on the response sheet.

⚠ Pre-Class Reading – Mandatory: All topics in today's MCQ are from the pre-class note. Students who have not read the note will not be able to answer the questions. Individual responsibility applies.

 IN-CLASS MCQ – Phase 1

Name: _____ Roll No.: _____

Date: _____ Time: _____

⚡ Flip Learning MCQ – Switching Characteristics and SCR Two-Transistor Model

Marking: Correct: +2 Wrong: $-\frac{2}{3}$ Not attempted: 0

🔑 Section A: Conceptual (Q1–Q5)

2 marks each

Q1. An ideal power semiconductor switch differs from a practical switch primarily in that the ideal switch:

- (a) Has a very low but non-zero ON-state voltage drop (b) Has zero ON-state voltage drop, zero OFF-state leakage current, and switches instantaneously with no switching loss (c) Switches instantaneously but still has a small conduction loss (d) Has no leakage current but has finite switching times

Q2. The turn-on time t_{on} of a power MOSFET is defined as:

- (a) The time for the drain current to rise from 0% to 100% of its ON-state value (b) The sum of the delay time t_d and the rise time t_r , where t_r is the time for current to rise from 10% to 90% of its ON-state value (c) The time for the gate voltage to reach its threshold value (d) The time for the device to enter saturation from the active region

Q3. In a practical power switch, the **storage time** t_s (which forms part of the turn-off time) is associated with:

- (a) The time for the device voltage to rise from 10% to 90% of the supply voltage (b) The time for the output capacitance to discharge through the load (c) The time required for excess minority carriers stored in the base/gate region to be swept out or recombined after the gate signal is removed (d) The time for the current to fall from 100% to 10% of its ON-state value

Q4. In the two-transistor model of an SCR, the cross-coupling between Q1 (PNP) and Q2 (NPN) is such that:

- (a) The emitter of Q1 is connected to the emitter of Q2 (b) The collector of Q1 drives the base of Q2, and the collector of Q2 drives the base of Q1, forming a positive feedback loop (c) The base of Q1 and the base of Q2 are both connected to the gate terminal (d) The collector of Q1 is connected to the collector of Q2 through the gate terminal

Q5. Once an SCR has been triggered into the ON state by a gate pulse, the gate pulse:

- (a) Must be maintained continuously to keep the SCR ON (b) Can be removed, since the regenerative feedback ($\alpha_1 + \alpha_2 \geq 1$) sustains the ON state as long as the anode current exceeds the holding current I_H (c) Can be removed only after the anode current reaches its maximum rated value (d) Must be reversed (negative gate pulse) to maintain the ON state

Section B: Analytical / Application (Q6–Q10) 2 marks each | Show brief working

Q6. A power IGBT switches at $f_s = 20$ kHz. Its turn-on time is $t_{on} = 500$ ns and turn-off time is $t_{off} = 1$ μ s. The OFF-state voltage is 400 V and the ON-state current is 20 A. Assuming linear rise and fall, the total average **switching power loss** is:

- (a) 24 W (b) 48 W (c) 60 W (d) 120 W

Q7. For the same IGBT in Q6, if the switching frequency is doubled to $f_s = 40$ kHz (all other parameters unchanged), the switching power loss becomes:

- (a) 24 W (b) 48 W (c) 60 W (d) 120 W

Q8. In the two-transistor model of an SCR, the anode current is given by $I_A = (I_G + I_{CBO1} + I_{CBO2})/(1 - \alpha_1 - \alpha_2)$. If $\alpha_1 = 0.45$, $\alpha_2 = 0.50$, $I_G = 5$ mA, and leakage currents are negligible, the anode current I_A is:

- (a) 10 mA (b) 50 mA (c) 100 mA (d) Theoretically unbounded (SCR latches ON)

Q9. An SCR has $\alpha_1 + \alpha_2 = 0.6$ at low anode voltages. As the anode-to-cathode voltage V_{AK} is increased without any gate signal, the device fires spontaneously when:

- (a) V_{AK} exceeds the forward voltage drop of 0.7 V (b) $\alpha_1 + \alpha_2$ reaches unity due to avalanche multiplication at the breakover voltage V_{BO} (c) The gate terminal receives a displacement current due to the rising V_{AK} (d) The anode current exceeds the latching current I_L

Q10. A power device has $t_{on} = 2$ μ s and $t_{off} = 3$ μ s. The **maximum** switching frequency $f_{s,max}$ at which the device can practically operate (assuming the switching period must be at least 10 times the total switching transition time) is approximately:

- (a) 10 kHz (b) 20 kHz (c) 50 kHz (d) 200 kHz

Total Marks: 20 Duration: 25 min Negative Marking: $-\frac{2}{3}$ per wrong

 Answer Key – For Faculty Reference Only (Not for Distribution)

Q	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Ans	(b)	(b)	(c)	(b)	(b)	(d)	(d)	(c)	(b)	(b)

1 Detailed Solutions – Faculty Reference

Section A: Conceptual (Q1–Q5)

Q1. An ideal switch has three defining properties: zero ON-state voltage drop (hence zero conduction loss), zero OFF-state leakage (hence zero blocking loss), and instantaneous switching (hence zero switching loss). Option (a) and (c) admit conduction loss; option (d) admits finite switching time. **Answer: (b)**

Q2. The turn-on time is $t_{on} = t_d + t_r$. The delay time t_d is from gate signal application to the point where drain current reaches 10% of its ON-state value (input capacitance charging). The rise time t_r is from 10% to 90% of the ON-state current. Option (a) uses 0%–100%; options (c) and (d) describe unrelated intervals. **Answer: (b)**

Q3. Storage time t_s is a phenomenon specific to minority-carrier (bipolar) devices: BJT, IGBT, and SCR. After the gate/base signal is removed, excess minority carriers stored in the base region must be swept out or recombine before the device can begin to turn off. During t_s , the current does not yet fall. This is why IGBTs and SCRs have longer turn-off times than power MOSFETs (which are majority-carrier devices with no t_s). Options (a) and (d) describe the voltage rise and current fall which happen after t_s . **Answer: (c)**

Q4. In the two-transistor model, the SCR is split into Q1 (PNP: Anode-N1-P2) and Q2 (NPN: N1-P2-Cathode). The collector of Q1 (P2 side) connects to the base of Q2 (P2 side); the collector of Q2 (N1 side) connects to the base of Q1 (N1 side). This forms a cross-coupled positive feedback loop – the defining feature that produces the regenerative (latching) turn-on. **Answer: (b)**

Q5. Once the regenerative condition $\alpha_1 + \alpha_2 \geq 1$ is satisfied, both Q1 and Q2 are in saturation and sustain each other through internal feedback. No external gate current is needed. The gate controls turn-on only, not the sustained ON state. To turn OFF, the anode current must be reduced below the holding current I_H through circuit action (commutation), not through the gate. **Answer: (b)**

Section B: Analytical (Q6–Q10)

Q6. Switching loss calculation:

$$E_{on} = \frac{1}{2} \times V_{off} \times I_{on} \times t_{on} = \frac{1}{2} \times 400 \times 20 \times 500 \times 10^{-9} = 2 \times 10^{-3} \text{ J} = 2 \text{ mJ}$$

$$E_{off} = \frac{1}{2} \times 400 \times 20 \times 1 \times 10^{-6} = 4 \times 10^{-3} \text{ J} = 4 \text{ mJ}$$

$$P_{sw} = (E_{on} + E_{off}) \times f_s = 6 \times 10^{-3} \times 20000 = 120 \text{ W}$$

Wait – rechecking: $(2 + 4) \times 10^{-3} \times 20 \times 10^3 = 6 \times 10^{-3} \times 20 \times 10^3 = 120 \text{ W}$. **Answer: (d)**

Instructor note: Update answer key – Q6 correct answer is (d), not (c). The distractor (c) 60 W corresponds to using only E_{off} or only half the correct f_s . Reprint answer key with Q6:(d).